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Multiple, Brief Maternal Separations in the Squirrel Monkey: Changes in Hormonal and Behavioral Responsiveness

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HENNESSY, M. B. *Multiple, brief maternal separations in the squirrel monkey: Changes in hormonal and behavioral responsiveness.* PHYSIOL BEHAV 36(2) 245–250, 1986.—Infant squirrel monkeys had their mother removed from the home cage for 2 hr on 80 occasions between 12 and 31 weeks of age. The initial separation elicited high levels of infant vocalizations. By the second observation session (Separation 14), no vocal response to separation was observed. Plasma cortisol levels were markedly elevated at the beginning of the separation series, and there was a significant decline in this response by Separation 28, which was the next separation during which cortisol was monitored. However, there was no further decline, so that significant cortisol elevations were observed throughout the remainder of the separation series. At 35 weeks, the infants were separated one additional time and their responses were compared to those of a previously nonseparated control group. The control group vocalized more, but the two groups exhibited equivalent cortisol elevations. Observations of time spent riding on the mother during undisturbed conditions indicated that both groups developed independence from the mother at about the normal rate. Overall, the data show that brief separation from the mother can activate the infants' pituitary-adrenal system even when the infant has been separated 80 times previously, no longer appears behaviorally responsive to separation, is almost 9 months of age, and exhibits normal signs of independence from the mother.

Multiple separation Maternal separation Pituitary-adrenal Plasma cortisol Vocalizations Squirrel monkey

THE infant squirrel monkey *(Saimiri sciureus)* exhibits a strong social relationship or attachment to its mother as shown by tests of recognition and preference, and by behavioral agitation upon separation [12, 14, 15]. Removal of the mother also triggers a physiological reaction associated with increased stress or arousal; that is, a sharp increase in pituitary-adrenal activity, as evidenced by rapid rises in plasma concentrations of cortisol [4].

Not only does brief separation evoke pituitary-adrenal activation in young *Saimiri,* but studies to date have found no habituation of this response. When exposed to six maternal separations of either 1 or 4 hr in length, infants responded with increases in cortisol levels during the final separation that were as great as those seen during the first removal of the mother [3,5]. A third study [13] found no change in the magnitude of the cortisol response across 20 separations that averaged 30 min in length. However, in each of these studies, the young *Saimiri* were separated in novel surroundings (Coe, personal communication re: [5]). It is known that in laboratory rodents the adrenocorticoid response to novelty alone can be quite resistant to habituation [10]. Thus, the repeated exposure to novelty could have contributed to the persistence of the cortisol response observed in these experiments. Another point that remains unclear is the relationship between cortisol and behavioral responses during repeated separations. For instance, in two of the studies cited above [3,5], there was a rapid reduction in vocalization rate across six separations, whereas Jordon *et al.* [13] observed a high and steady rate of vocalizations during the period from Separation 8–20 that they monitored their animals. Additional data relating changes in behavior to those in cortisol during repeated separations in familiar surroundings are needed to clarify this issue.

The present study examined the cortisol and behavioral responses of infant squirrel monkeys during an extended series of brief maternal separations (80, 2-hr separations) that did not involve exposure to novelty. During separation, the number of high-pitched vocalizations emitted by infants was monitored; this is probably the most frequently employed measure of the infant squirrel monkey's reaction to separation procedures (e.g., [5,8]). To help distinguish between changes in responsiveness due to repeated separations and those due to maturation, the subjects were separated one additional time and their responses were compared to those of a previously nonseparated control group of the same age. Finally, to provide a measure of the development of independence in separated and nonseparated infants, the time spent riding on the mother was monitored weekly under undisturbed conditions. The decline with age in the amount of time infants spend on the mother is commonly used to assess the development of independence in squirrel monkeys and other primates [11, 22, 23].

METHOD

Animals

Eleven infant *Saimiri sciureus* of the Bolivian variety served as experimental animals. Two males and three females comprised the separated (S) group; the remaining three males and three females served as nonseparated (N) controls. Each infant was housed with its mother in a standard primate cage (74×64×38 cm). The cages were situated in a temperature-controlled indoor colony room that contained no animals other than those used in this study. All cages were aligned along one wall and no animal could view monkeys in other cages. The monkeys were maintained on a 12-hr light/dark cycle (lights on at 0630), and had ad lib access to New World monkey chow and water. A fruit-flavored drink (Tang®) containing Vitamin C was provided one day per week. In order to monitor the general health of the animals, each S and N infant was weighed once weekly. This procedure involved removing the infant from the home cage, weighing it while it was in a carrying cage, and then quickly returning it to the home cage.

Separation Procedure

Beginning at Week 12 of age, and continuing through Week 31, each S infant was separated from its mother for 2 hr per day, four days per week, for a total of 80 separations. Each separation consisted of removing the mother from the home cage. She was then placed alone in a cage in a remote room of the animal housing facility to prevent vocal communication between mother and infant during the separation period. At the end of 2 hr, the mother was returned to the home cage. Separations were initiated between 1300 and 1500 hr.

At 35 weeks of age, S as well as N infants were separated permanently from their mothers. The procedure for this final separation was identical to that for the earlier multiple separations of S animals, except that the mother was not returned to the home cage at the end of 2 hr.

Behavioral Observations

Vocalizations. During the multiple separations, the number of high-pitched vocalizations emitted by S infants was monitored every 13 separations (i.e., Separations 1, 14, 27, 40, 53, 66 and 79). Observation sessions (15 min) were conducted during three different periods: preseparation, separation and reunion. Preseparation observations were made prior to any disturbance on a day of the same week as the designated separation, at approximately the same time of day. Separation observations were made Min 60–74 of the designated separation period. Reunion observations were made on the day of separation, immediately following the return of the mother to the home cage.

Vocalizations emitted by S and N monkeys during the final separation were also monitored. Each monkey was observed for 15 min on two occasions: at the conclusion of the first hr of separation (as had been done for S monkeys during the earlier separations) and at 24 hr.

Infant independence. To assess the development of infant independence, each S and N monkey was observed under undisturbed conditions for one, 15-min session per week during Weeks 8–31. These observations were made during morning hr (0930–1200 hr), and were, therefore, completed prior to any separation to be imposed that day. During these observations, the observer noted every 30 sec, on the signal, whether the infant was riding on the mother in either the dorsal-ventral or ventral-ventral positions. All behavioral observations were made by experienced observers, from an observation booth, through one-way glass.

Blood Sample Collection and Cortisol Determination

During the multiple separations, blood samples from S monkeys were collected for analysis of plasma cortisol levels every 26 separations, beginning with the second (i.e., Separations 2, 28, 54 and 80). These samples were taken at the end of the 2-hr separation period. The first blood sample was collected during Separation 2, rather than Separation 1, because it was not possible to collect blood when reunion behavior was being observed. Blood samples from S monkeys were collected for determination of resting cortisol concentrations on Week 11 (the week prior to the initiation of the separation procedure) and on Week 32 (the week following the last of the 80 separations). The N monkeys had their blood sampled the same number of times and at the same ages as did the S animals. This was done to control for any effects that the blood sampling procedure might have on other measures, e.g., the behavioral and cortisol responses to the final separation. Blood samples from N monkeys were generally discarded, although those collected on Week 32 were used, as will be described below.

Blood samples were taken from S and N animals 2 hr and 24 hr following the final separation at 35 weeks of age. The 24-hr sample was drawn immediately following the behavioral observation scheduled at this time. Blood samples collected from S and N monkeys under undisturbed conditions at 32 weeks of age were used for a resting-level comparison.

To collect a blood sample, the animal was rapidly anesthetized in a glass jar containing ethyl-ether-soaked cotton. Approximately 0.8 ml of blood was then taken directly from the heart, using a heparinized syringe and injection needle (25 gauge×1.6 cm). Blood samples were collected within about 3 min of the time the animal was taken from the cage. Thus, based on available evidence from laboratory rodents [6,7], the samples were collected before the disturbance of the blood-sampling procedure itself could have produced any appreciable increase in circulating cortisol levels. Young squirrel monkeys regain consciousness, exhibit coordinated behavior and can be returned to their mothers within minutes of this procedure.

Blood samples were centrifuged to separate plasma which was then frozen until assayed. Plasma aliquots were assayed in triplicate using a radioimmunoassay kit from New England Nuclear (^{3}H). The column chromatography step was omitted because pilot data indicated that it was unnecessary.

RESULTS

Multiple Separations

For analysis of vocalization scores during multiple separations, a parametric analysis of variance (ANOVA) was not employed because, in a number of cells, 3 or 4 of the 5 individual scores were zeros. Instead, Friedman nonparametric ANOVAS by ranks were used to compare the number of vocalizations emitted during the preseparation, separation and reunion periods for each observed separation. A significant difference among observation periods was observed for the first ($\chi_r^2=7.90$, $p<0.05$) but not subsequent separations. During the first separation, each monkey vocalized much more frequently during the separation period

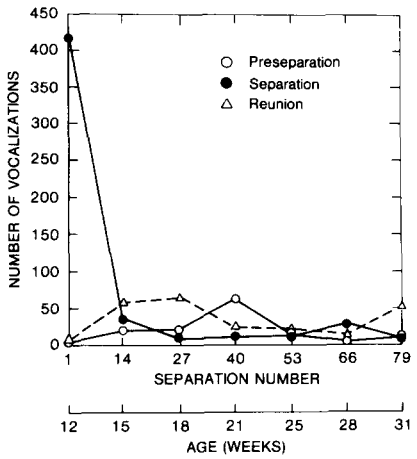


FIG. 1. Mean number of vocalizations (per 15 min) emitted by infants in the preseparation, separation and reunion periods of every 13th separation of the multiple separation series. Also shown is infant age at the time of each indicated separation.

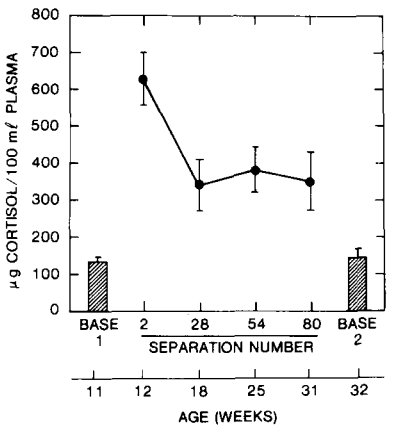


FIG. 2. The line graph illustrates the mean plasma cortisol concentrations of infants 2 hr following removal of the mother during Separations 2, 28, 54 and 80. The bars depict the infants' mean resting cortisol levels. The vertical lines indicate the standard errors of the means. Also shown is infant age at the time of each blood sample collection.

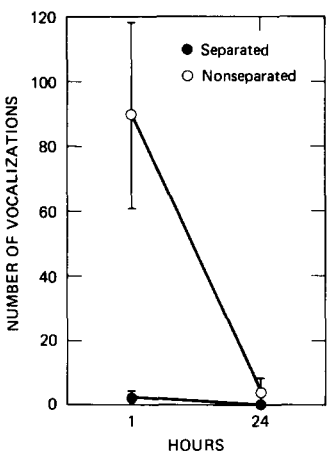


FIG. 3. Mean number of vocalizations (per 15 min) emitted by the previously separated and the previously nonseparated monkeys 1 and 24 hr following removal of the mother at 35 weeks of age. Vertical bars indicate the standard errors of the means.

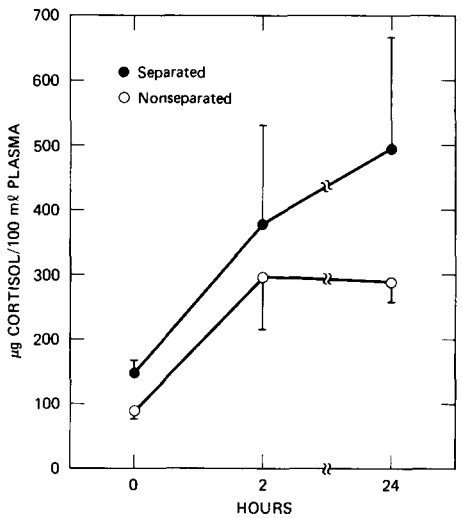


FIG. 4. Mean plasma cortisol concentrations of the previously separated and the previously nonseparated monkeys 2 and 24 hr following removal of the mother at 35 weeks of age. Mean resting cortisol levels (0 hr) at 32 weeks are also shown. Vertical bars indicate the standard errors of the means.

than any monkey did during either of the other two periods (lowest separation score=207; highest preseparation and reunion scores=19 and 33, respectively; see Fig. 1).

For plasma cortisol, a 1-way repeated-measures ANOVA showed that there was a significant change in cortisol concentrations across the separation series, $F(3,12)=14.03$, $p<0.01$, Fig. 2. Newman-Keuls post-hoc comparisons revealed that cortisol levels were significantly higher during the second separation then they were during Separations 28, 54 or 80 ($ps<0.01$). There was no difference among the values obtained during these latter three separations. Resting cortisol levels from Weeks 11 and 32 did not differ ($t<1$). The mean of the resting cortisol levels observed at these two times was then compared to the mean level observed during Separations 28, 54 and 80. This analysis showed that cortisol levels during these separations were significantly elevated over resting levels, $t(4)=3.08$, $p<0.05$.

Final Separation

The mean numbers of vocalizations emitted by S and N monkeys following the permanent removal of the mother at Week 35 are shown in Fig. 3. Parametric analyses were not used for these data because most scores in three of the four cells were zeros. A Mann-Whitney U test showed that N infants vocalized more than S infants at the 1-hr time point, $U=0$, $p<0.01$. At 24 hr, only one of the 11 subjects emitted any vocalizations whatsoever.

Figure 4 illustrates the plasma cortisol levels of S and N

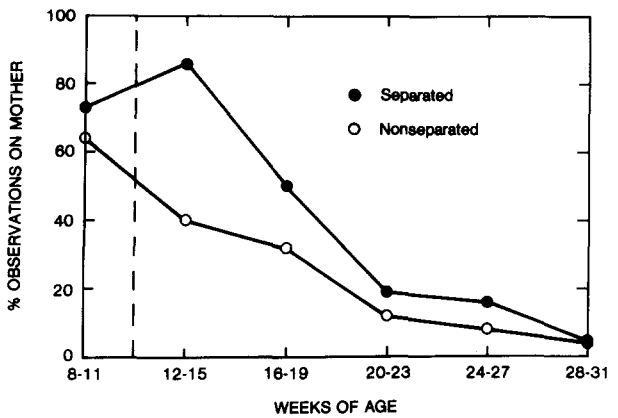


FIG. 5. The number of times that infants of both groups were observed riding on the mother in either the dorsal-ventral or ventral-ventral position during undisturbed observations from Week 8–31. The data were grouped into 4-week blocks prior to analysis, and are expressed here as a percentage of the total number of instances (120) the animals were observed in each block (every 30 sec for 15 min each week). The 8–11 week observations were made prior to the separation group's first separation.

monkeys 2 and 24 hr following the final separation, as well as their resting levels (0 hr) obtained at Week 32. Because of heterogeneous variances, the data were subjected to a log transformation, and were then analyzed with a 2 (Rearing Groups) × 3 (Timepoints) ANOVA. A significant main effect for Timepoint was obtained; $F(2,18)=14.45$, $p<0.01$, indicating that separation evoked an increase in plasma cortisol levels. There was no difference between Rearing Groups in resting and/or separation-induced cortisol levels, as indicated by nonsignificant Rearing Groups and Rearing Groups × Timepoints effects.

Independence

The undisturbed observations made during morning hours were grouped into 4-week blocks. The first block (Weeks 8–11) was comprised of observations occurring before the initial separation of S infants, and was therefore analyzed separately. No difference was found in the number of times the S and N subjects were observed riding on the mother during this first block ($t<1$). Following the initial separation, differences between the groups emerged. A 2 (Rearing Groups) × 5 (Blocks) ANOVA, with Blocks treated as a repeated measure, yielded significant main effects for Rearing Groups, $F(1,9)=6.43$, $p<0.05$, and for Blocks, $F(4,36)=12.18$, $p<0.01$. As illustrated in Fig. 5, S monkeys were observed on their mothers more frequently than were their N counterparts. Although the Rearing Groups × Blocks interaction was not significant, the difference between the S and N groups was most apparent during the earlier blocks. The effect of Blocks reflects the development of infant independence, i.e., the decline with age in the amount of time spent on the mother. Although S monkeys were observed on the mother more than N monkeys were in general, animals of both groups achieved independence at about the same time, spending relatively little time on the mother from Week 20–23 onward.

DISCUSSION

Infants responded to the first separation of the multiple separation series with a nearly continual emission of high-pitched vocalizations. However, there was a precipitous decline in vocalizations with repeated separations, so that by the second observation session (Separation 14), there was no longer any increase in vocalizing observed during the separation period. At Week 35, the previously nonseparated monkeys vocalized significantly more 1 hr following separation than did the previously separated animals. This indicates that the reduction in vocalizations across separations in S monkeys was at least partially due to habituation. Furthermore, the much higher rate of vocalizing observed in S monkeys during their first separation at Week 12 as compared to N monkeys during their first separation at Week 35 indicates that there was an age-related decline in responsiveness to separation. This could also have contributed to the reduction in vocalizations observed between Separations 1 and 14 in the S monkeys. These results corroborate earlier findings by Coe *et al.* [3,5], which together indicate that vocal reactions typically disipate rapidly during repeated, brief maternal separations in the squirrel monkey.

During Separation 2, infant cortisol concentrations showed a 4.5-fold increase over resting levels. By Separation 28, there was a significant decline in the cortisol response. However, from the 28th to the 80th separation, the magnitude of the cortisol response remained unchanged; separation continued to elicit significant corticoid elevations during this period. At 35 weeks of age, there was no effect of earlier separation history on pituitary-adrenal responsiveness. That is, the S monkeys, which had been separated on 80 previous occasions, and the N monkeys, which had not undergone the multiple separation regimen, exhibited equivalent plasma cortisol concentrations following the final separation. This indicates that the waning of the cortisol response might be totally accounted for by an age-related reduction in responsiveness to the separation procedure. Still, the possibility of some effect of habituation cannot be ruled out. That is, the decline in the cortisol response may have occurred at an earlier age than it would have if the monkeys had been separated less frequently. Further, without examining the entire time course of the cortisol response across repeated separations, it is impossible to ensure that there was no change in response dynamics (e.g., rate of return to resting level). It should also be pointed out that the stimulus for pituitary-adrenal activation in this experiment was almost certainly the absence of the mother, rather than the disturbance involved in capturing the animals, because earlier studies have found such disturbance to have minimal or no effect on cortisol levels [9, 16, 19].

During observations made under undisturbed conditions, S monkeys were on the mother more often than were N monkeys once the multiple separation series had begun, but not prior to the initiation of this procedure. This finding could have been the result of either: (a) a separation-induced increase in infant dependency during the ages at which separations were being imposed, (b) compensatory clinging resulting from the recent removal of the clingable object (mother), or (c) some degree of exhaustion as a result of increased exertion (e.g., vocalizing) during separation the previous afternoon (particularly during the earlier separations). It is not possible to decide among these alternative explanations from the data. Of greater concern here, monkeys of both groups showed a clear decline with age in time

spent on the mother. By 20–23 weeks of age, S as well as N monkeys were observed on the mother during less than 20% of the observations. The timing of this decline agrees with earlier observations of *Saimiri* maintained in social groups [22,23], and suggests that the S and N animals of this study developed physical independence from the mother at about the normal rate for laboratory-reared *Saimiri*.

In the current study, there was a striking persistence of the cortisol response. Although there was an initial, partial decline in the magnitude of this response during the early separations, what is more notable is the lack of a further reduction. Infants continued to respond to separation with increased plasma cortisol levels even when they had been separated 80 times previously, when the vocalization response had long ago waned, and when the monkeys were nearly 9 months of age and appeared to be quite independent as indicated by the measure of time spent on the mother. (The infants were surely nutritionally independent by this time; infant *Saimiri* are typically weaned by about 6 months in the laboratory.) Moreover, since the infants were always separated in the familiar home cage, repeated exposure to novelty could not have contributed to the persistence of the cortisol response. Clearly, removal of the mother is not only a potent stimulus for pituitary-adrenal activation, but also one that can be extremely resistant to habituation.

In general, there is still much debate concerning the course of pituitary-adrenal responsiveness during repeated presentations of a stimulus (stressor). In rodents, habituation, no change, and sensitization of adrenocorticoid responses have all been found in similar testing paradigms [10,21]. At present, the conditions that determine how corticoid responses will change during subsequent presentations of a stressor have not been fully resolved. One factor that surely plays a part is the number of stimulus presentations—presumably the corticoid response to most effective stimuli will eventually habituate if the stimulus is presented often enough. Yet most studies employ relatively few stimulus presentations. The results reported here show that the cortisol response of the squirrel monkey to the nonnociceptive stressor of brief maternal separation need not habituate completely even after 80 presentations. Indeed, there may have been no habituation at all, though it is not possible to unequivocally determine this from the present results.

The persistence of the cortisol response in this experiment also suggests the possibility of conditioning. That is, the removal of the mother may have served as an unconditioned stimulus that was repeatedly paired with certain other events, such as opening of the cage door and initial capture of the mother-infant pair. These latter events might have then become conditioned stimuli which could have elicited the cortisol response in the absence of actual maternal separation. This sort of neuroendocrine conditioning has not been studied in the context of mother-infant separation, but it is known that pituitary-adrenal responsiveness can be readily conditioned to environmental stimuli in other situations [1,17].

The age at which removal of the mother, even initial removal, would no longer evoke increased pituitary-adrenal activity remains an open question. Bernstein and Dobrofsky [2] found that juvenile/adolescent pigtailed macaques (2–5 years old) still displayed behavioral signs of distress when the mother was removed from the social group. It would be interesting to determine if squirrel monkeys that were several years old would show a neuroendocrine reaction to the removal of their mother.

A number of investigators have studied the effects of multiple separations on the behavior of the rhesus macaque [18, 20, 24, 25, 26]. As has been discussed previously [3,13], differences both in focus and procedures (particularly the emphasis on prolonged vs. brief separations) make it difficult to draw direct comparisons between the rhesus studies and experiments such as the present one, of brief separation in the squirrel monkey. However, it should be noted that declines in vocalizations during the course of repeated maternal separations have been observed in rhesus infants in the study that most closely resembles the present one [26]. One implication the present results would have for future investigations of multiple separation in any primate species is the caution that, during repeated separations, an infant may appear calm behaviorally but still be displaying a significant neuroendocrine "stress response". What consequences such continual neuroendocrine activation might have for the young primate at some later time remains to be determined.

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REFERENCES

1. Ader, R. Conditioned adrenocortical steroid elevations in the rat. *J Comp Physiol Psychol* **90:** 1156–1163, 1976.
2. Bernstein, I. S. and M. Dobrofsky. Compensatory social responses of older pigtailed monkeys to maternal separation. *Dev Psychobiol* **14:** 163–168, 1981.
3. Coe, C. L., J. C. Glass, S. G. Wiener and S. Levine. Behavioral, but not physiological, adaptation to repeated separation in mother and infant primates. *Psychoneuroendocrinology* **8:** 401–409, 1983.
4. Coe, C. L., S. P. Mendoza, W. P. Smotherman and S. Levine. Mother-infant attachment in the squirrel monkey: adrenal responses to separation. *Behav Biol* **22:** 256–263, 1978.
5. Coe, C. L., S. G. Wiener, L. T. Rosenberg and S. Levine. Physiological consequences of maternal separation and loss in the squirrel monkey. In: *Handbook of Squirrel Monkey Research*, edited by L. A. Rosenblum and C. L. Coe. New York: Plenum Publishing Company, 1985, pp. 127–148.
6. Coover, G. D., J. P. Heybach, J. Lenz and J. Miller. Corticosterone "basal levels" and response to ether anesthesia in rats on a water deprivation regimen. *Physiol Behav* **22:** 653–656, 1979.
7. Davidson, J. M., L. E. Jones and S. Levine. Feedback regulation of adrenocorticotropin secretion in "basal" and "stress" conditions: acute and chronic effects of intrahypothalamic corticoid implantation. *Endocrinology* **82:** 655–663, 1968.
8. Hennessy, M. B. Effects of surrogate-rearing on the infant squirrel monkey. In: *Handbook of Squirrel Monkey Research*, edited by L. A. Rosenblum and C. L. Coe. New York: Plenum Publishing Company, 1985, pp. 149–168.
9. Hennessy, M. B., J. N. Kaplan, S. P. Mendoza, E. L. Lowe and S. Levine. Separation distress and attachment in surrogate-reared squirrel monkeys. *Physiol Behav* **23:** 1017–1023, 1979.
10. Hennessy, M. B. and S. Levine. Effects of various habituation procedures on pituitary-adrenal responsiveness in the mouse. *Physiol Behav* **18:** 799–802, 1977.

11. Hinde, R. A. and Y. Spencer-Booth. Effects of brief separation from mother on rhesus monkeys. *Science* **173:** 111–118, 1971.
12. Jones, B. C. and D. L. Clark. Mother-infant separation in squirrel monkeys living in a group. *Dev Psychobiol* **6:** 259–269, 1973.
13. Jordan, T. C., C. L. Coe, J. Patterson and S. Levine. Predictability and coping with separation in infant squirrel monkeys. *Behav Neurosci* **98:** 556–560, 1984.
14. Kaplan, J. The effects of separation and reunion on the behavior of mother and infant squirrel monkeys. *Dev Psychobiol* **3:** 43–52, 1970.
15. Kaplan, J. N., D. Cubicciotti, III and W. K. Redican. Olfactory discrimination of squirrel monkey mothers by their infants. *Dev Psychobiol* **10:** 447–454, 1977.
16. Levine, S., C. L. Coe, W. P. Smotherman and J. N. Kaplan. Prolonged cortisol elevation in the infant squirrel monkey after reunion with mother. *Physiol Behav* **20:** 7–10, 1978.
17. Levine, S. and G. D. Coover. Environmental control of suppression of the pituitary-adrenal system. *Physiol Behav* **17:** 35–37, 1976.
18. McKinney, W. T., Jr., S. J. Suomi and H. F. Harlow. Repetitive peer separations of juvenile-age rhesus monkeys. *Arch Gen Psychiatry* **27:** 200–203, 1972.
19. Mendoza, S. P., W. P. Smotherman, M. T. Miner, J. Kaplan and S. Levine. Pituitary-adrenal response to separation in mother and infant squirrel monkeys. *Dev Psychobiol* **11:** 169–175, 1978.
20. Mineka, S., S. J. Suomi and R. DeLizio. Multiple separations in adolescent monkeys: An opponent-process interpretation. *J Exp Psychol Gen* **110:** 56–85, 1981.
21. Pfister, H. P. The glucocorticosterone response to novelty as a psychological stressor. *Physiol Behav* **23:** 649–652, 1979.
22. Rosenblum, L. A. Mother-infant relations and early behavioral development in the squirrel monkey. In: *The Squirrel Monkey*, edited by L. A. Rosenblum and R. W. Cooper. New York: Academic Press, 1968, pp. 207–233.
23. Rosenblum, L. A. and C. L. Coe. The influence of social structure on squirrel monkey socialization. In: *Primate Bio-Social Development*, edited by S. Chevalier-Skolnikoff and F. Poirier. New York: Garland Publishing Company, 1977, pp. 479–499.
24. Spencer-Booth, Y. and R. A. Hinde. Effects of 6 days separation from mother on 18- to 32-week-old rhesus monkeys. *Anim Behav* **19:** 174–191, 1971.
25. Suomi, S. J., H. Harlow and C. J. Domek. Effect of repetitive infant-infant separation of young monkeys. *J Abnorm Psychol* **76:** 161–172, 1970.
26. Suomi, S. J., S. Mineka and R. D. DeLizio. Short- and long-term effects of repetitive mother-infant separations on social development in rhesus monkeys. *Dev Psychol* **19:** 770–786, 1983.